

Phase 2: audit → re-measure → re-test

the question changed from 'which batch is different?' to 'which numbers can we defend?'

Every marker re-derived under the frozen Batch_3-fitted recipe; every claim re-graded

Sorted into a scorecard, not a verdict — MARKER_SCORECARD.md + AUDIT_LEDGER.md

Tier A — validated descriptors (describe, don't discriminate)

Tier B — candidate associations (real in crops, unprovable as material)

Tier C — model-derived + correlation layer (fresh, corrected)

Tier X — demoted/withdrawn (documented, not deleted)

9 defects reproduced and repaired on the way (contact denominators,

BH step-up, NaN→p=0, channel masks, crack direction, QC weight leak...)

Deliverable = an evidence ledger — every number traced to file + test

What survives: shared structure, one corrected ordering

Tier A — all three batches look like the same kind of material

marker	B1 / B2 / B3	meaning
candidate-Si fraction	0.0595 / 0.0614 / 0.0603	~6% Si everywhere — kills 'B1 excess Si'
resolved pore fraction	0.104 / 0.108 / 0.109	same resolved porosity
pore orientation S_dir	0.585 / 0.563 / 0.573	near-horizontal pores in ALL batches
pore contact (corrected)	0.483 / 0.378 / 0.354	B1 most connected — ordering, not activity
Si enclosed share	0.720 / 0.778 / 0.781	consistent: B1 least enclosed
low-coverage Si	0.867 / 0.882 / 0.906	87–91% blind spot — sub-resolution pores
patchiness (hotspot)	3.75 / 3.78 / 4.29	equally patchy, no mega-cluster

Constraint: 0 markers clear the declared minimum-detectable-difference (v2 batch_differences: zero survivors)

The confound, quantified — and the one survivor

predeclared benchmark: can any marker set classify batch?

protocol	best honest set	envelope	acq. signature	majority
leave-one-image-out	51.6%	59.3%	61.3% ← wins	54.8%
leave-group-out	41.9%	54.2%	38.7%	54.8% ← ties

The imaging-session signature alone beats the material envelope

the pipeline sees the photography session better than the material

Corrected blind-test bookkeeping: 0/2 confirmed (3e→B1 was B2,

fn→B2 was B1 — both flagged
cant_tell_session_matched in advance)

Lone BH survivor: pores_per_mpx — B1>B3 +34.5 (p=0.0005),
pore_d50 agrees (more, smaller pores in B1) — but B1-B3 share ONE

matched image: real association, untestable confound

External 76-98% scores: documented as not ours

What was fixed or withdrawn

Tier X — retracted in the record, not hidden; archive/ARCHIVE.md

FFT 'vertical texture' — withdrawn: measured frame aspect ratio;
replaced by validated S-parameter orientation (synthetic + null tested)

Crack-like metric — was selecting horizontal objects labeled vertical;
two definitions still disagree → report per-definition or drop

Accessible-Si denominator bug — exemplar 0.524 → 0.371;
renamed: resolved-pore-contact proxy, not activity

'Uncertain-bright $\geq$ Si brightness' — false in B1 after mask fix
(InLens 1.395 < 1.459); true only in B3

Stats bugs — BH elementwise error, NaN→p=0, QC weight leak
(redundancy fit on pooled batches) — all tables regenerated green

D10 particle size — grazing cuts fake small particles:
demoted to 2-D profile diagnostic; tile-'jackknife' renamed tile SE

Correlation layer + fresh DFN — and the honest headline

Tier C: model-derived indicators, not measurements

layer	result	reading
S2 correlation	pore len x/y anisotropy ≈ 1.5 , all batches	independent confirmation of horizontal pores
PCF $g(r)$ vs CSR	peak 5.4–7.2 at $\sim 0.75\ \mu\text{m}$, above envelope	Si clustered beyond random — everywhere
DFN v2 stress	B1–B3 = $-3.0\ \text{MPa}$ (CI $-6.02, -0.02$)	meaningful-looking BUT inside assumption spread (IQR 3.9)
DFN capacity/swelling	below declared thresholds	no model-level difference

Same amounts (~6% Si, ~10% pores),
horizontally-elongated pores, Si clustered beyond random), one real
contact ordering in B1's favor — but the pipeline classifies the session
better than the material: no batch/manufacturing difference claimable

What remains: validated descriptors + frozen honest ruler + preregistered
blind-test protocol for the six incoming images + a ledger that shows its work